COMPETENCY STANDARD

FOR

3D PRINTING TECHNOLOGY

(Light Engineering Sector)

Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance

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INTRODUCTION

The Skills for Industry Competitiveness and Innovation Program (SICIP) has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and ensure that the required skills are aligned with industry requirements.

The Competency Standard also suggests integration of YouTube or similar digital platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.

The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competence and its corresponding Elements.

OVERVIEW:

A **Competency Standard** is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.

Competency standard:

- provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
- enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies
- encourages the development and delivery of flexible training that suits individual and industry requirements
- encourages learning and assessment in a work-related environment which leads to verifiable workplace outcomes.

Competency Standard has been developed by a working group comprised of occupation-specific experts from the industry/institution and relevant consultants of SICIP.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

- Reference to Industry Sector, Occupational Title and Occupational Description
- Unit code
- Unit title
- Unit descriptor
- Unit of Competency
- Elements and performance criteria
- Variables and range statement
- Evidence guides

Together all the parts of a Unit of Competency:

- Describe a work activity
- Guide the assessor in determining whether the candidate is competent.

Identification and validation of units of competency and elements for each occupation were made by experts from various Light Engineering companies in industry consultative workshops.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competence:

- An overview of all Units of Competence for the occupation and their corresponding duration required for completion of training.
- The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

Units & Elements at a Glance:
Generic Competencies (20 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-LE-3DPT-01-G	Apply Occupational Health and Safety (OHS) Practices in the Workplace	1. Identify OHS policies and procedures 2. Practice personal health and safety procedures 3. Report hazards and risks 4. Respond to emergencies	10
SICIP-LE-3DPT-02-G	Operate in a Self-Directed Team	1. Identify team goals and work processes 2. Communicate and cooperate with team members. 3. Work as a team member. 4. Solve problems as a team member	10
		Total Hour	**20**

Sector Specific Competencies (10 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-LE-3DPT-01-S	Apply Green Practices	1. Interpret green concepts 2. Minimize resource use in the workplace 3. Implement waste management practices	10
		Total Hour	**10**

Occupation Specific Competencies (270 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (hours)
SICIP-LE-3DPT-01-O	Understand the fundamentals of 3D Printing technology	1. Interpret overview of 3D Printing Technology 2. Identify FDM Printing Technology 3. Identify SLA Printing Technology 4. Recognize the role of slicer software in 3D printing	10
SICIP-LE-3DPT-02-O	Perform 3D modelling using CAD Software	1. Install CAD software 2. Carry out 2D sketching for design creation 3. Design 3D models using feature and surface tools 4. Assemble 3D models	85
SICIP-LE-3DPT-03-O	Perform Fused Deposition Modelling (FDM) Printing	1. Prepare the FDM printer for operation 2. Configure print settings and parameters using slicer 3. Monitor the 3D printing process 4. Perform Post-processing of Printed Parts	70
SICIP-LE-3DPT-04-O	Perform Masked Stereolithography (MSLA) Printing	1. Prepare the MSLA printer for operation 2. Configure print settings and parameters using slicer 3. Monitor the 3D printing process 4. Perform Post-processing of Printed Parts	50
SICIP-LE-3DPT-05-O	Implement Design Principles for Additive Manufacturing (DfAM)	1. Understand the principles and advantages of DfAM 2. Apply DfAM Principles to Enhance Product Design	10
SICIP-LE-3DPT-06-O	Perform Prototyping using 3D scanning technology	1. Conduct 3D scanning of physical objects 2. Perform post-processing of scanned files 3. Print prototype	35

SICIP-LE-3DPT-07-O	Perform Basic Maintenance of 3D Printer	1. Identify and troubleshoot 3D printer issues 2. Conduct preventive maintenance of the 3D printer	10
		Total Hour	**270**

COMPETENCY STANDARD: 3D PRINTING TECHNOLOGY

A. The Generic Competencies

Unit of Competency:	Nominal Duration:	Unit Code:
APPLY OCCUPATIONAL HEALTH AND SAFETY (OHS) PRACTICES IN THE WORKPLACE	10 hrs.	SICIP-LE-3DPT-01-G
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to apply occupational health and safety (OHS) practices in the workplace. It specifically includes the tasks of identifying OHS policies and procedures, practicing personal health and safety procedures, reporting hazards and risks, and responding to emergencies.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify OHS policies and procedures	1.1 **<u>OHS policies</u>** and safe operating procedures are read and understood. 1.2 Safety signs and symbols are identified and followed. 1.3 Emergency response, evacuation procedures and other contingency measures are determined.
2. Practice personal health and safety procedures	2.1 OHS policies and procedures are followed and practiced. 2.2 **<u>Personal Protective Equipment (PPE)</u>** is selected and used. 2.3 Personal health, hygiene and safety procedures are practiced.
3. Report hazards and risks	3.1 **<u>Hazards and risks</u>** are identified, assessed and controlled. 3.2 Incidents arising from hazards and risks are reported to authority. 3.3 Corrective actions are implemented to correct unsafe conditions in the workplace.
4. Respond to emergencies	4.1 Alarms and warning devices are responded. 4.2 **<u>Emergency response plans and procedures</u>** are implemented. 4.3 **<u>First aid procedure</u>** is applied during emergency situations.

Range of Variables

Variable	Range May include but not limited to:
1. OHS policies	1.1 International/ Local OHS requirements 1.2 Fire Safety Rules and Regulations 1.3 Industry Guidelines

2. Personal Protective Equipment (PPE)	2.1 Apron 2.2 Gloves 2.3 Safety shoes 2.4 Helmet 2.5 Face mask 2.6 Goggles and safety glasses 2.7 Ear plugs 2.8 Sun block 2.9 Chemical/Gas masks
3. Hazards and risks	3.1 Chemical hazards 3.2 Biological hazards 3.3 Physical Hazards 3.3.1 Machine hazards 3.3.2 Materials hazards 3.3.3 Tools and Equipment hazards
4. Emergency response plans and procedures	4.1 Firefighting procedures 4.2 Earthquake response procedures 4.3 Evacuation procedures 4.4 Medical and first-aid
5. First aid procedure	5.1 Washing of open wound 5.2 Washing chemically infected area 5.3 Applying bandage 5.4 Applying CPR (Cardiopulmonary Resuscitation) 5.5 Taking appropriate medicine

Curricular Content Guide:

1. Underpinning Knowledge	1.1 OHS workplace policies and procedures 1.2 Work safety procedures 1.3 Emergency procedures 1.3.1 Firefighting 1.3.2 Earthquake response 1.3.3 Explosion response 1.3.4 Accident response 1.4 Types of hazards (biological, chemical and physical) and their effects 1.5 Types of PPE and their uses 1.6 Personal hygiene practices 1.7 OHS awareness
2. Underpinning Skills	2.1 Identifying OHS policies and procedures 2.2 Applying personal work safety practices 2.3 Reporting hazards and risks 2.4 Responding to emergency procedures 2.5 Maintaining physical well-being in the workplace 2.6 Using firefighting accessories and fire extinguishers 2.7 Applying basic first aid procedures
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities

	3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Learning/operational manual 4.3 Monitoring tools/equipment (CCTV, sensor, alarm, etc.) 4.4 PPEs 4.5 Firefighting equipment 4.6 Emergency response manual 4.7 Emergency Systems (Fire alarms, emergency exits, etc.) 4.8 First aid kits

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 applied OHS policies and procedures 1.2 selected and used the PPE 1.3 practiced personal health and safety procedures 1.4 reported incidents arising from hazards and risks to the authority 1.5 implemented plans and procedures to respond emergency 1.6 applied basic first aid procedure
2. Methods of Assessment	Methods of assessment should include: 2.1 Written test 2.2 Practical demonstration 2.3 Oral question 2.4 Portfolio (optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training 3.2 Assessment should be done by a certified assessor or occupation-specific industry expert.

Unit of Competency: **OPERATE IN A SELF-DIRECTED TEAM**	Nominal Duration: 10 hrs.	Unit Code: SICIP-LE-3DPT-02-G
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to operate in a self-directed team. It specifically includes the tasks of identifying team goals and work processes, communicating and cooperating with team members, working and solving problems as a team member.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify team goals and work processes	1.1 Team goals and collaborative decision-making processes are identified. 1.2 Roles and responsibilities of team members are identified. 1.3 Relationships within team and with other workers are identified.
2. Communicate and cooperate with team members.	2.1 Effective interpersonal skills are used to interact with team members and to contribute to activities and objectives. 2.2 Formal and informal **forms of communication** are used effectively to support team achievement. 2.3 Diversity in character is respected and valued in team functioning. 2.4 Views and opinions of other team members are understood, valued and practiced. 2.5 Workplace terminology is used correctly to assist communication.
3. Work as a team member	3.1 Duties, responsibilities, authorities, objectives and task requirements are identified and clarified with team. 3.2 Tasks are performed in accordance with organizational and team requirements, and workplace procedures & practices. 3.3 Team member's support to other members is appreciated and valued. 3.4 Agreed reporting lines are followed using standard operating procedure.
4. Solve problems as a team member	4.1 Current and potential problems faced by team are identified. 4.2 A solution to the problem is identified. 4.3 Problems are solved effectively and the outcome of the implemented solution is evaluated.

Range of Variables

Variable	Range May Include but not limited to:
1. Forms of communication	1.1 Agenda 1.2 Simple reports such as progress and incident reports 1.3 Job sheets 1.4 Operational manuals 1.5 Brochures and promotional material 1.6 Visual and graphic materials 1.7 Standards 1.8 OSH information 1.9 Signs

Curricular Content Guide:

1. Underpinning Knowledge	1.1 Team goals and collaborative decision-making processes 1.2 Roles and responsibilities of team members 1.3 Relationships within team and with other workers 1.4 Effective formal and informal forms of communication 1.5 Value of diversity in team functioning 1.6 Correct use of workplace terminology 1.7 Team's duties, responsibilities, authorities, objectives and task requirements 1.8 Support mechanism to other members of team to ensure achievements of goals 1.9 Effective problem-solving methods and evaluation of outcomes
2. Underpinning Skills	2.1 Identifying team goals and collaborative decision-making processes 2.2 Identifying roles and responsibilities of team members 2.3 Using effective interpersonal skills to interact with team members and to contribute to activities and objectives 2.4 Using formal and informal forms of communication 2.5 Understanding and valuing views and opinions of other team members 2.6 Performing tasks in accordance with organizational and team requirements and workplace procedures & practices 2.7 Identifying current and potential problems faced by a team 2.8 Identifying solutions to the problem 2.9 Solving problems effectively and evaluating the outcome of the implemented solution
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness

	3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Work books 4.3 Learning manuals 4.4 Flexible organizational policies 4.5 Communication platforms 4.6 Performance tracking tools

Assessment Evidence Guide:

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified team goals and work processes 1.2 described roles and responsibilities of team members 1.3 communicated and cooperated with team members 1.4 explained formal and informal forms of communication 1.5 worked as an active team member 1.6 solved problems effectively as a team member
2. Methods of Assessment	Methods of assessment should include: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training 3.2 Assessment should be done by a certified assessor or occupation-specific industry expert

B. Sector Specific Competencies

Unit of Competency: **APPLY GREEN PRACTICES**	Nominal Duration: 10 hrs.	Unit Code: SICIP-LE-3DPT-01-S
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to apply green practices. It specifically includes the tasks of interpreting green concepts, minimizing resource use in the workplace and implementing waste management practices.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret green concepts	1.1 **<u>Principles of green practices</u>** are explained. 1.2 **<u>Sources of environmental impacts</u>** during light engineering activities are identified. 1.3 Work activities contributing to environmental degradation, improper disposal of materials and excessive energy use are explained. 1.4 **<u>Ways of mitigating environmental impacts</u>** in light engineering sector are explained.
2. Minimize resource use in the workplace	2.1 Water, energy, and raw material consumption are documented. 2.2 **<u>Recyclable and non-recyclable items</u>** are identified. 2.3 Procedures to reduce resource consumption are implemented. 2.4 Sustainable alternatives to fossil-based energy resources are explored and applied.
3. Implement waste management practices	3.1 **<u>Different types of wastes</u>** are identified. 3.2 Hazardous waste is disposed of according to environmental regulations. 3.3 Green habits to reduce wastes in personal and professional life are practiced.

Range of Variables

Variable	**Range** May include but not limited to:
1. Principles of green practices	1.1 Reducing energy consumption 1.2 6R for waste management 1.2.1 Refuse 1.2.2 Reduce 1.2.3 Reuse 1.2.4 Recycle 1.2.5 Recover 1.2.6 Repair 1.3 Use of sustainable materials with low environmental impact

	1.4 Recycling materials 1.5 Sustainable transportation
2. Sources of environmental impacts	2.1 Air pollution 2.1.1 Dust generation 2.1.2 Emission from machinery and vehicles 2.2 Water pollution 2.2.1 Debris and sediments 2.2.2 Chemical's leakage 2.3 Noise pollution 2.4 Waste generation 2.4.1 Waste 2.4.2 Non-recyclable and hazardous materials 2.5 Resource Depletion 2.5.1 Excessive use of raw materials 2.5.2 Non-renewable material uses like use of fossil fuel, steel cement etc.
3. Ways of mitigating environmental impacts	3.1 Utilizing Energy-Efficient Equipment 3.2 Adopting Renewable Energy Sources 3.3 Implementing Site Protection Measures 3.4 Using Reusable Materials 3.5 Choosing Sustainable Materials 3.6 Using Noise-Reducing Equipment and scheduling work appropriately 3.7 Optimizing Logistics and Delivery
4. Recyclable and non-recyclable items	4.1 Recyclable items 4.1.1 Metal 4.1.2 Wood 4.1.3 Brick 4.1.4 Concrete 4.2 Non-recyclable items 4.2.1 Paints and coatings 4.2.2 Contaminated materials like lead paint, asbestos 4.2.3 Treated wood
5. Different types of waste	5.1 Wood waste 5.2 Metal waste 5.3 Plastic waste 5.4 Electrical system waste 5.5 Gypsum board waste 5.6 Packaging waste 5.7 Organic waste 5.8 Chemical West 5.9 Fuel/Oil west

Curricular Content Guide

1. Underpinning	1.1 Principles of green practices in Light Engineering sector

Knowledge	1.2 Key terms and symbols related to environmental sustainability in Light Engineering drawings 1.3 Sources of environmental impacts in Light Engineering 1.4 Ways of mitigating environmental impacts 1.5 Waste management practices 1.6 Sustainable alternatives to fossil-based energy resources 1.7 Personal and workplace habits for reducing environmental impact
2. Underpinning Skills	2.1 Identifying environmental impacts in Light Engineering activities 2.2 Identifying recyclable and non-recyclable items 2.3 Applying methods to minimize resource use in the workplace 2.4 Implementing waste management practices 2.5 Applying procedures to safely disposal of hazardous materials 2.6 Documenting water, energy and material consumption in the workplace 2.7 Practicing green habits to reduce waste in both personal and professional life
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Light engineering hand tools and power tools 4.3 Environmental compliance (Government regulations, standards and inspections) 4.4 Green technologies 4.5 Monitoring system (software, sensor) 4.6 Learning manuals 4.7 Waster segregation bins

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 explained the principles of green practices in Light Engineering 1.2 identified sources of environmental impacts 1.3 implemented procedures to minimize resource use (water, energy, raw materials) 1.4 disposed of hazardous waste in compliance with safety

	and environmental standards 1.5 practiced green habits to reduce waste in both personal and professional life
2. Methods of Assessment	Methods of assessment should include: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

C. Occupation-specific Competencies

Unit of Competency: UNDERSTAND THE FUNDAMENTALS OF 3D PRINTING TECHNOLOGY	Nominal Duration: 10 hrs.	Unit Code: SICIP-LE-3DPT-01-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to understand the fundamentals of 3D printing technology. It specifically includes the tasks of interpreting overview of 3D printing technology, identifying FDM printing technology, identifying SLA printing technology, and recognizing the role of slicer software in 3D printing.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret overview of 3D printing technology	1.1 Principles of 3D printing technology are understood and explained. 1.2 Differences between additive manufacturing and subtractive manufacturing are understood. 1.3 **3D printing methods** are identified and described. 1.4 Uses and applications of 3D printing technology are understood. 1.5 Evolution of 3D printing technology is traced from its inception to the present. 1.6 **3D printing procedures** are identified. 1.7 Impact of 3D printing on various industries is interpreted.
2. Identify FDM printing technology	2.1 **FDM printing terminology** is defined and applied in the context of 3D printing. 2.2 Various **types of FDM printing** are identified. 2.3 General working principle of FDM printing is explained. 2.4 Role of FDM printing is described. 2.5 **FDM printing materials** are identified and explained. 2.6 Advantages and limitations of FDM printing technology are understood.
3. Identify SLA printing technology	3.1 **SLA printing terminology** is defined. 3.2 Different **types of SLA printing** are identified. 3.3 General working principle of SLA printing is described. 3.4 **Materials used in SLA printing** are identified. 3.5 **Post-processing steps** of SLA printing are explained. 3.6 Advantages and limitations of SLA printing technology are understood.
4. Recognize the role of slicer software in 3D printing	4.1 Function and importance of slicer software is recognized. 4.2 3D model orientation is identified within slicer software. 4.3 3D model manipulation is understood within the slicer software. 4.4 **Printing parameters** are identified.

Range of Variables

Variable	Range May include but not limited to:
1. 3D printing methods	1.1 Binder Jetting 1.2 Material Jetting 1.3 Directed Energy Deposition 1.4 Sheet Lamination 1.5 Material Extrusion 1.6 Vat Photopolymerization 1.7 Powder Bed Fusion
2. 3D printing procedures	2.1 Design the 3D model 2.2 Export and prepare the file 2.3 Slice the model 2.4 Set up the printer 2.5 Print the part 2.6 Post-processing 2.7 Inspection and application
3. FDM printing terminology	3.1 Fused Deposition Modeling (FDM) 3.2 Filament 3.3 Hotend 3.4 Extruder 3.5 Nozzle 3.6 Build Platform (or Bed) 3.7 G-code 3.8 Layer Hight 3.9 Infill 3.10 Print speed 3.11 Travel speed 3.12 Shell 3.13 Printing or nozzle temperature 3.14 Bed temperature 3.15 Retraction
4. Types of FDM printing	4.1 Cartesian 4.2 Delta 4.3 CoreXY 4.4 SCARA 4.5 Belt
5. FDM printing materials	5.1 PLA (Polylactic Acid) 5.2 ABS (Acrylonitrile Butadiene Styrene) 5.3 PETG (Polyethylene Terephthalate Glycol) 5.4 Nylon 5.5 PC (Polycarbonate) 5.6 TPU (Thermoplastic Polyurethane) 5.7 PP (Polypropylene) 5.8 Glass fiber filled filament 5.9 Carbon fiber filled filament 5.10 Metal filled filament

	5.11 Wood filled filament
6. SLA printing terminology	6.1 Stereolithography (SLA) 6.2 Photopolymer Resin 6.3 Vat Photopolymerization 6.4 UV Laser/light 6.5 Resin tank or vat 6.6 LCD screen 6.7 Layer thickness 6.8 Normal exposer time 6.9 Bottom exposer time 6.10 Normal layer 6.11 Bottom layer 6.12 Off time 6.13 Z lift Hight 6.14 Z lift speed 6.15 Retract speed 6.16 Anti-alias
7. Types of SLA printing	7.1 SLA (Stereolithography) 7.2 Masked Stereolithography (MSLA) 7.3 Digital Light Processing (DLP) 7.4 Continuous Liquid Interface Production (CLIP)
8. Materials used in SLA printing	8.1 Standard Resins 8.2 Engineering Resins 8.3 Castable Resins 8.4 Dental Resins 8.5 Clear Resins 8.6 Rubber like Resins
9. Post-processing steps	9.1 Washing 9.2 Curing
10. Printing parameter	10.1 Print speed 10.2 Layer height 10.3 Temperature 10.4 Infill density 10.5 Support settings 10.6 Retraction settings

Curricular Content Guide

1. Underpinning knowledge	1.1 Principles and applications of 3D printing technology 1.2 3D printing methods 1.3 FDM and SLA printing terminologies 1.4 General working principles of FDM and SLA printings 1.5 Post-processing steps of SLA printing 1.6 Function and importance of slicer software
2. Underpinning skills	2.1 Identifying 3D printing methods 2.2 Identifying 3D printing procedures 2.3 Applying FDM printing terminology in the context of 3D printing.

	2.4 Identifying various types of FDM printing 2.5 Identifying FDM printing materials 2.6 Identifying different types of SLA printing 2.7 Identifying materials used in SLA printing 2.8 Identifying 3D model orientation within slicer software 2.9 Identifying printing parameters
3. Underpinning attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Learning materials 4.3 3D Printers (Access to FDM, SLA) 4.4 Computers & Software (CAD, slicing software) 4.5 Internet & learning platforms 4.6 Drawings and sketches

Assessment Evidence Guide

1. Critical aspects of competency	Assessment must evidence that the candidate: 1.1 described principles of 3D printing technology 1.2 interpreted general working principles of FDM and SLA printings 1.3 described function and importance of slicer software 1.4 identified 3D printing methods and procedures 1.5 applied FDM printing terminology in the context of 3D printing 1.6 applied SLA printing terminology in the context of 3D printing 1.7 identified 3D model orientation within slicer software
2. Methods of assessment	Methods of assessment should include: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency:	Nominal Duration:	Unit Code:

PERFORM 3D MODELLING USING CAD SOFTWARE	85 hrs.	SICIP-LE-3DPT-02-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to perform 3D modelling using CAD software. It specifically includes the tasks of installing CAD software, carrying out 2D sketching for design creation, designing 3D models using feature and surface tools, and assembling 3D models.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Install CAD software	1.1 System requirements for SolidWorks software are identified. 1.2 Installation process of SolidWorks software is followed. 1.3 Licensing and activation keys are entered. 1.4 Software is activated to ensure legitimate usage. 1.5 Required plugins or extensions are installed and configured within the SolidWorks software. 1.6 Functionality of the SolidWorks software is tested by opening a sample project or 3D model.
2. Carry out 2D sketching for design creation	2.1 User interface is recognized within SolidWorks software. 2.2 Drawing workspace is set up within SolidWorks software. 2.3 Basic **<u>2D sketching tools</u>** are utilized to create the foundational elements of the design. 2.4 Geometrical constraints are applied using **<u>relation tools</u>** to define relationships between 2D entities. 2.5 Dimensions and annotations are added as per job requirement. 2.6 Sketch errors or inconsistencies are identified and corrected. 2.7 2D sketch is saved and exported in the appropriate format for further processing.
3. Design 3D models using feature and surface tools	3.1 **<u>Feature tools</u>** are utilized as per requirement. 3.2 **<u>Surface tools</u>** are used to achieve the desired complex geometry. 3.3 Geometric constraints and relationships are used to control model features. 3.4 3D model is modified and refined. 3.5 Model is reviewed and validated by checking for errors, interferences, or design flaws. 3.6 3D model is saved in a compatible **<u>file format</u>** for use in 3D printing or further design iterations.
4. Assemble 3D models	4.1 Individual 3D models are correctly selected and imported.

	4.2 3D models are organized and modified using appropriate **assembly tools.** 4.3 Assembly constraints are applied using appropriate **mating tools**. 4.4 Interference or collision checks are performed. 4.5 Movable components are simulated to verify the correct range of motion. 4.6 Assembly is saved in a compatible format for further processing or production.

Range of Variables

Variable	**Range** May include but not limited to:
1. 2D sketching tools	1.1 Line 1.2 Circle 1.3 Point 1.4 Arc 1.5 Rectangle 1.6 Polygon 1.7 Spline 1.8 Slot 1.9 Pattern 1.10 Trim 1.11 Mirror 1.12 Offset
2. Relation tools	2.1 Parallel 2.2 Horizontal 2.3 Vertical 2.4 Concentric 2.5 Fix 2.6 Equal 2.7 Tangent 2.8 Coincide 2.9 Colinear
3. Feature tools	3.1 Extrude boss/cut 3.2 Revolve boss/cut 3.3 Swept boss/cut 3.4 Loft boss/cut 3.5 Fillet/chamfer 3.6 Shell and rib 3.7 Mirror and pattern 3.8 Hole wizard
4. Surface tools	4.1 Extrude 4.2 Boundary 4.3 Loft 4.4 Swept 4.5 Filled

	4.6 Offset 4.7 Knit 4.8 Trip 4.9 Flaten 4.10 Thicken
5. File format	5.1 .STL 5.2 .OBJ 5.3 .FBX 5.4 .IGES 5.5 .STEP 5.6 .X3D 5.7 .PLY 5.8 .3MF 5.9 .AMF 5.10 .DAE
6. Assembly tools	6.1 Insert component 6.2 Move or rotate component 6.3 Edit component 6.4 Pattern component
7. Mating tools	7.1 Concentric 7.2 Parallel 7.3 Perpendicular 7.4 Angle 7.5 Profile center 7.6 Width mate 7.7 Angle limit 7.8 Gear 7.9 Slot

Curricular Content Guide

1. Underpinning knowledge	1.1 Installation process of SolidWorks software 1.2 User interface within SolidWorks software 1.3 Feature tools and surface tools 1.4 Geometric constraints and relationships
2. Underpinning skills	2.1 Identifying system requirements for SolidWorks software 2.2 Installing and configuring required plugins or extensions within the SolidWorks software 2.3 Utilizing basic 2D sketching tools and creating foundational elements of the design 2.4 Applying geometrical constraints using relation tools to define relationships between 2D entities 2.5 Exporting 2D sketch in the appropriate format for further processing 2.6 Designing 3D model using feature and surface tools 2.7 Saving 3D model in a compatible file format for use in 3D printing or further design iterations 2.8 Organizing and modifying 3D models using appropriate

	assembly tools 2.9 Applying assembly constraints using appropriate mating tools
3. Underpinning attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Hardware **(**Computer [CPU, GPU, RAM, SSD], high-resolution monitors) 4.3 Software (CAD software [AutoCAD, SolidWorks, etc.]) 4.4 Network and IT (Internet [for cloud-based CAD tools]) 4.5 Learning materials 4.6 Tools & equipment 4.7 Job specifications and work instructions

Assessment Evidence Guide

1. Critical aspects of competency	Assessment must evidence that the candidate: 1.1 explained installation process of SolidWorks software 1.2 described feature tools and surface tools 1.3 installed and configured required plugins or extensions within the SolidWorks software 1.4 utilized basic 2D sketching tools and created foundational elements of the design 1.5 exported 2D sketch in the appropriate format for further processing 1.6 designed 3D model using feature and surface tools 1.7 organized and modified 3D models using appropriate assembly tool
2. Methods of assessment	Methods of assessment should include: 2.1 Written Test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/ training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **PERFORM FUSED DEPOSITION MODELLING (FDM) PRINTING**	Nominal Duration: 70 hrs.	Unit Code: SICIP-LE-3DPT-03-O
Unit Descriptor: This unit of competency covers the knowledge, skills and attitudes required to perform fused deposition modelling (FDM) printing. It specifically includes the tasks of preparing the FDM printer for operation, configuring print settings and parameters using slicer, monitoring the 3D printing process and performing post-processing of printed parts.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare the FDM printer for operation	1.1 Safety protocols are followed for preparing and operating the printer. 1.2 Printer bed levelling is performed ensuring that the print bed is leveled and aligned. 1.3 Z-offset calibration is performed as required. 1.4 Filament is loaded into the extruder as per requirement. 1.5 **Calibration test** related printer and filament is carried out. 1.6 Pre-print checks are completed ensuring secure and functioning.
2. Configure print settings and parameters using slicer	2.1 Printer settings are configured according to the material specifications and print requirements. 2.2 3D model is imported within in the slicer. 2.3 **Model manipulation** is performed. 2.4 Pre-print settings are configured in the slicer. 2.5 **Print parameters** are adjusted in the slicer. 2.6 Print preview is checked, ensuring that the slicing parameters produce a valid print path. 2.7 Printing program or G-code is generated for printing. 2.8 Print program is transferred to the printer.
3. Monitor the 3D printing process	3.1 Print progress is observed, ensuring that the first layers attached with the printer bed. 3.2 Print quality is checked during printing process. 3.3 Printer parameters are checked to ensure they remain within the required range for optimal printing. 3.4 Errors and issues are identified and addressed ensuring minimal disruption to the printing process.
4. Perform post-processing of printed parts	4.1 Printed parts are removed from the print bed ensuring that the part is not damaged during removal. 4.2 Excess support structures are removed using appropriate tools like pliers or knives. 4.3 Dimensional accuracy is checked the required specifications.

	4.4 Post-curing or cooling is applied if necessary, ensuring that the part is fully hardened or cooled. 4.5 Surface finish is improved by sanding or filing the part, ensuring smooth edges and surfaces. 4.6 Paint, coating, or adhesion is applied as required according to the design specifications. 4.7 Final inspection is conducted, ensuring that the part is functional, visually appealing, and free from defects.

Range of Variables

Variable	**Range** May include but not limited to:
1. Calibration test	1.1 Temperature test 1.2 Retraction test 1.3 Bridge test 1.4 Overhang test 1.5 Scale test 1.6 Clearance test 1.7 E-step calibration 1.8 Acceleration calibration 1.9 Jerk calibration
2. Model manipulation	2.1 Move 2.2 Scale 2.3 Rotate 2.4 Copy 2.5 Support blocker
3. Print parameters	3.1 Layer height 3.2 Wall 3.3 Print temperature 3.4 Bed temperature 3.5 Infill density 3.6 Infill pattern 3.7 Print speed 3.8 Flow rate 3.9 Bed adhesion 3.10 Support structures

Curricular Content Guide

1. Underpinning knowledge	1.1 Safety protocols for preparing and operating FDM printer 1.2 Print parameters 1.3 Printing program or G-code
2. Underpinning skills	2.1 Performing FDM printer bed levelling ensuring that the print bed is leveled and aligned 2.2 Carrying out calibration test related printer and filament 2.3 Configuring printer settings and pre-print settings 2.4 Performing model manipulation 2.5 Adjusting print parameters in the slicer

	2.6 Generating printing program or G-code for printing 2.7 Monitoring the 3D printing process, including print quality 2.8 Checking dimensional accuracy as per required specifications 2.9 Performing post-processing of printed parts along with final inspection
3. Underpinning attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, sub-ordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Computer/laptop 4.3 Software (slicing, CAD) 4.4 Equipment & hardware (FDM 3D Printer, heated bed & nozzle system, maintenance tools) 4.5 Specifications and work instructions 4.6 Learning materials

Assessment Evidence Guide

1. Critical aspects of competency	Assessment must evidence that the candidate: 1.1 described safety protocols for preparing and operating FDM printer 1.2 explained print parameters 1.3 performed FDM printer bed levelling 1.4 configured printer settings and pre-print settings 1.5 adjusted print parameters in the slicer 1.6 generated printing program or G-code for printing 1.7 performed post-processing of printed parts and final inspection
2. Methods of assessment	Methods of assessment should include: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/ training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **PERFORM MASKED STEREO LITHOGRAPHY (MSLA) PRINTING**	**Nominal Duration:** 50 hrs.	**Unit Code:** SICIP-LE-3DPT-04-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to perform masked stereo lithography (MSLA) printing. It specifically includes the tasks of preparing the MSLA printer for operation, configuring print settings and parameters using slicer, monitoring the 3D printing process and performing post-processing of printed parts.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **<u>bold and underlined</u>** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare the MSLA printer for operation	1.1 Safety checks are performed with proper ventilation and protective gear. 1.2 Resin tank, build platform, and light source are installed and made functioning. 1.3 Printer bed levelling is performed and printer home position is set. 1.4 UV light source is calibrated for accurate curing of the resin during each layer. 1.5 Resin is loaded into the resin tank. 1.6 Calibration test is carried out for optimal exposure time.
2. Configure print settings and parameters using slicer	2.1 Printer settings are configured to match the specific printer and material being used. 2.2 3D model is imported within in the slicer. 2.3 **<u>Model manipulation</u>** is performed. 2.4 **<u>Print parameters</u>** are set in the slicer ensuring the settings match the print job requirements. 2.5 Support structures are defined in the slicer. 2.6 Print preview is checked and optimized for smooth printing. 2.7 Printing program is generated from the slicer to execute the print job. 2.8 Print program is transferred to the printer.
3. Monitor the 3D printing process	3.1 Print progress is observed ensuring the layers are properly cured. 3.2 Print quality is checked during printing process. 3.3 Printer parameters are checked to ensure they remain within the required range for optimal printing. 3.4 Errors and issues are identified and addressed ensuring minimal disruption to the printing process.
4. Perform post-processing of printed parts	4.1 Printed parts are removed from the print bed ensuring no damage occurs during removal.

	4.2 Printed parts are washed using washing agent. 4.3 Support structures are removed using appropriate tools such as pliers or knives. 4.4 Post-curing is performed for optimal strength and durability. 4.5 Dimensional accuracy is checked with the required specifications. 4.6 Surface finishing is applied and enhanced the final appearance. 4.7 Painting or coating is applied if required.

Range of Variables

Variable	**Range** May include but not limited to:
1. Model manipulation	1.1 Move 1.2 Scale 1.3 Rotate 1.4 Copy 1.5 Layout
2. Print parameters	2.1 Layer thickness 2.2 Normal exposer time 2.3 Bottom exposer time 2.4 Normal layer 2.5 Bottom layer 2.6 Transition layer setting 2.7 Off time 2.8 Z lift Hight 2.9 Z lift speed 2.10 Retract speed 2.11 Anti-alias 2.12 Infill 2.13 Hollow 2.14 Punch 2.15 Model split 2.16 Text paste

Curricular Content Guide

1. Underpinning knowledge	1.1 Safety checks with proper ventilation and protective gear 1.2 Printing program
2. Underpinning skills	2.1 Preparing the MSLA printer for operation 2.2 Setting print parameters in the slicer (ensuring the settings match the print job requirements) 2.3 Configuring printer settings and parameters using slicer (to match the specific printer and materials being used) 2.4 Generating printing program from the slicer to execute printing 2.5 Monitoring the 3D printing process, including print quality 2.6 Checking dimensional accuracy as per required

	specifications 2.7 Performing post-processing of printed parts along with final inspection
3. Underpinning attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in workplace 3.8 Good relationships with peers, sub-ordinates and seniors in workplace
4. Resource implications	The Following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Computer/laptop 4.3 Equipment (MSLA printer, curing & cleaning stations, ventilation) 4.4 Materials (Resin, IPA, gloves, FEP films) 4.5 Software (Slicer, CAD design tools) 4.6 Learning manual 4.7 Job specifications and work instructions

Assessment Evidence Guide

1. Critical aspects of competency	Assessment must evidence that the candidate: 1.1 explained safety checks with proper ventilation and protective gear 1.2 described printing program 1.3 prepared the MSLA printer for operation 1.4 configured printer settings and parameters using slicer 1.5 generated printing program from the slicer 1.6 performed post-processing of printed parts along with final inspection
2. Methods of assessment	Methods of assessment should include: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/ training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **IMPLEMENT DESIGN PRINCIPLES FOR ADDITIVE MANUFACTURING (DfAM)**	**Nominal Duration:** 10 hrs.	**Unit Code:** SICIP-LE-3DPT-05-O
Unit Descriptor: This unit of competency covers the knowledge, skills and attitudes required to implement design principles for additive manufacturing (DfAM). It specifically includes the tasks of understanding the principles and advantages of DfAM and applying DfAM principles to enhance product design.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **<u>bold and underlined</u>** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Understand the principles and advantages of DfAM	1.1 **<u>Principles of Design for Additive Manufacturing (DfAM)</u>** are understood. 1.2 Benefits of DfAM is interpreted. 1.3 DfAM based on process constraints and machine considerations are understood. 1.4 DfAM based on design constraints and geometric is understood. 1.5 **<u>Advance DfAM methods</u>** are identified.
2. Apply DfAM Principles to Enhance Product Design	2.1 3D model geometry is analyzed for compatibility with the selected 3D printing process. 2.2 DfAM principles are applied based on the analysis to optimize design for material efficiency, structural integrity, and printability. 2.3 Design is iterated and optimized to reduce material usage, weight, and improve manufacturability. 2.4 Optimized model is printed to verify functionality and performance against design specifications.

Range of Variables

Variable	Range May include but not limited to:
1. Principles of Design for Additive Manufacturing (DfAM)	1.1 Features size 1.2 Part orientation, overhangs and support 1.3 Support structures 1.4 Hole orientation 1.5 Model splitting 1.6 Material properties 1.7 Part consolidation 1.8 Lattice structure
2. Advance DfAM methods	2.1 Topology optimization 2.2 Generative design

Curricular Content Guide

1. Underpinning knowledge	1.1 Principles of Design for Additive Manufacturing (DfAM) 1.2 Benefits of DfAM
2. Underpinning skills	2.1 Identifying advance DfAM methods 2.2 Analyzing 3D model geometry for compatibility with the selected 3D printing process 2.3 Applying DfAM principles based on the analysis to optimize design for material efficiency, structural integrity and printability 2.4 Printing optimized model to verify functionality and performance against design specifications
3. Underpinning attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, subordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Computer/laptop 4.3 Learning manual 4.4 Software (Advanced CAD, simulation tools, slicing tools) 4.5 Hardware (Access to AM printers, post-processing machines, prototyping space) 4.6 Materials (Specialized AM materials, material qualification) 4.7 Specifications and work instructions

Assessment Evidence Guide

1. Critical aspects of competency	Assessment must evidence that the candidate: 1.1 described principles of Design for Additive Manufacturing (DfAM) 1.2 interpreted benefits of DfAM 1.3 analyzed 3D model geometry for compatibility with the selected 3D printing process 1.4 applied DfAM principles based on the analysis to optimize design for material efficiency, structural integrity and printability
2. Methods of assessment	Methods of assessment should include: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/ training center or in an actual or simulated

	work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **PERFORM PROTOTYPING USING 3D SCANNING TECHNOLOGY**	**Nominal Duration:** 35 hrs.	**Unit Code:** SICIP-LE-3DPT-06-O
Unit Descriptor: This unit of competency covers the knowledge, skills and attitudes required to perform prototyping using 3D scanning technology. It specifically includes the tasks of conducting 3D scanning of physical objects, performing post-processing of scanned files and printing prototype		

Elements and Performance Criteria
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Conduct 3D scanning of physical objects	1.1 **<u>3D scanning technology</u>** is defined. 1.2 Scanning software is installed and licensed ensuring the software set up and activate. 1.3 3D scanner is installed and integrated with the software. 1.4 Calibration and settings of the 3D scanner are performed. 1.5 3D scanning of the product is carried out, ensuring that the object is scanned from multiple angles. 1.6 Processing, refining and optimizing scanned data is performed.
2. Perform post-processing of scanned files	2.1 Geomagic Design X is defined. 2.2 Geomagic Design X software is installed for post-processing of the scanned data. 2.3 Scanned data is imported from scanning software. 2.4 Region segmentation and alignment of scanned data are performed. 2.5 **<u>CAD primitives</u>** are extracted from the scanned data into usable geometric shapes for CAD modeling. 2.6 Solid tools, surface tools, and auto surfacing are used for design manipulation and design finalization. 2.7 Files are saved in appropriate file format for slicer software.
3. Print prototype	3.1 Post-processed scanned file is imported to slicer. 3.2 Printer is prepared and loaded with correct material. 3.3 Print settings are configured in the slicing software. 3.4 Printing program is generated from the slicer software. 3.5 Print program is transferred to the printer. 3.6 Print progress is monitored, ensuring that the model is printing correctly and addressing any issues. 3.7 Prototype is completed including dimensional accuracy and surface finish.

Range of Variables

Variable	**Range** May include but not limited to:
1. 3D scanning technology	1.1 LASER triangulation 3D scanning technology 1.2 Structured light 3D scanning technology 1.3 Photogrammetry 1.4 Contact-based 3D scanning technology 1.5 Laser pulse-based 3D scanning technology 1.6 Laser phase-shift 3D scanning technology
2. CAD primitives	2.1 Box or cube 2.2 Cylinder 2.3 Sphere 2.4 Cone 2.5 Torus 2.6 Pyramid

Curricular Content Guide

1. Underpinning knowledge	1.1 3D scanning technology 1.2 Geomagic Design X software
2. Underpinning skills	2.1 Installing and licensing scanning software 2.2 Installing and integrating 3D scanner with the software 2.3 Carrying out 3D scanning of the product (object is scanned from multiple angles) 2.4 Installing Geomagic Design X software for post-processing of the scanned data 2.5 Configuring print settings in the slicing software 2.6 Generating printing program from the slicer software 2.7 Monitoring print progress including quality 2.8 Completing prototype including dimensional accuracy and surface finish
3. Underpinning attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, subordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Computer/laptop 4.3 Hardware (3D scanner, high-performance PC, calibration tools) 4.4 Software (Scanning, mesh editing, CAD/CAE software) 4.5 Learning materials 4.6 Specifications and work instructions

Assessment Evidence Guide

1. Critical aspects of competency	Assessment must evidence that the candidate: 1.1 interpreted 3D scanning technology 1.2 installed and integrated 3D scanner with the scanning software 1.3 carried out 3D scanning of the product 1.4 installed Geomagic Design X software for post-processing of the scanned data 1.5 performed post-processing of scanned files 1.6 completed and printed prototype
2. Methods of assessment	Methods of assessment should include: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **PERFORM BASIC MAINTENANCE OF 3D PRINTER**	Nominal Duration: 10 hrs.	Unit Code: SICIP-LE-3DPT-07-O
Unit Descriptor: This unit of competency covers the knowledge, skills and attitudes required to perform basic maintenance of 3D printer. It specifically includes the tasks of identifying and troubleshooting 3D printer issues, and conducting preventive maintenance of the 3D printer.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Troubleshoot 3D printer issues	1.1 **<u>Common 3D printer issues</u>** are identified. 1.2 Filament issues are diagnosed and resolved by cleaning the extruder or changing the filament. 1.3 Printer calibration is checked and aligned with the printing bed. 1.4 **<u>Electrical or mechanical issues</u>** are diagnosed and resolved. 1.5 Software or G-code issues are identified ensuring that the correct slicing settings. 1.6 Printer communication problems are diagnosed and resolved.
2. Conduct preventive maintenance of the 3D printer	2.1 **<u>Daily maintenance</u>** is performed in proper working condition. 2.2 **<u>Periodic maintenance tasks</u>** are scheduled and performed. 2.3 Software updates and firmware are checked regularly, ensuring that the printer's software is up to date.

Range of Variables

Variable	Range May include but not limited to:
1. Common 3D printer issues	1.1 Filament jams 1.2 Inconsistent extrusion 1.3 Layer misalignment 1.4 Poor bed adhesion 1.5 Warping 1.6 Stringing 1.7 Under-extrusion
2. Electrical or mechanical issues	2.1 Faulty wiring 2.2 Motor malfunction 2.3 Overheating 2.4 Timing belt wear 2.5 Bent lead screw

	2.6 T-nut wear
3. Daily maintenance	3.1 Nozzle cleaning 3.2 Bed cleaning 3.3 Filament drying 3.4 Apply lubrication 3.5 Resin vat cleaning 3.6 Belt tension 3.7 Nozzle Z offset 3.8 Bed levelling
4. Periodic maintenance tasks	4.1 Nozzle maintenance 4.2 Print bed maintenance 4.3 Belt and pulley maintenance 4.4 Lubrication 4.5 LCD screen 4.6 Vat FEP film 4.7 Moving Parts 4.8 Cables and connectors 4.9 Cooling Fans

Curricular Content Guide

1. Underpinning knowledge	1.1 Common 3D printer issues 1.2 Schedule of periodic maintenance tasks
2. Underpinning skills	2.1 Identifying software or G-code issues ensuring the correct slicing settings 2.2 Diagnosing and resolving electrical or mechanical issues 2.3 Diagnosing and resolving printer communication problems 2.4 Performing daily maintenance in proper working condition 2.5 Scheduling and performing periodic maintenance tasks
3. Underpinning attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincerity and honesty to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Good relationships with peers, subordinates and seniors in the workplace
4. Resource implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Computer/laptop 4.3 Tools and equipment (basic maintenance tools, cleaning supplies, spare parts) 4.4 Software and firmware (firmware updates, diagnostic tools) 4.5 Learning materials 4.6 Specifications and work instructions

Assessment Evidence Guide

1. Critical aspects of competency	Assessment must evidence that the candidate: 1.1 interpreted common 3D printer issues 1.2 described schedule of periodic maintenance tasks 1.3 identified software or G-code issues ensuring the correct slicing settings 1.4 diagnosed and resolved electrical/mechanical and printer communication problems 1.5 performed daily maintenance in proper working condition
2. Methods of assessment	Methods of assessment should include: 2.1 Written test 2.2 Practical demonstration 2.3 Oral questioning 2.4 Portfolio (optional)
3. Context of assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

END OF COMPETENCY STANDARD

Workshop/Lab Facility Standard

Course Name:	3D Printing Technology
Number of Trainees:	25

Course-wise Training Space:

- Computer Lab : 400 sft
- Workshop : 700-800 sft

Major Training Equipment and Training Facilities:

S.N.	Major Training Equipment and facilities	Required facilities
1.	Computer/Laptops	01
2.	Printer	01
3.	Multimedia projector	01
4.	Internet connectivity with power source	01
5.	3D Printers (FDM and MSLA)	04
6.	Laptops/Computers (for modeling and printer control)	25
7.	3D Printing Filaments	05
8.	Heat beds (for better print adhesion)	04
9.	3D Printer Maintenance Tools	02
10.	3D Modeling Software Licenses (SolidWorks)	25
11.	3D Printer Calibration Tools and gauge	02
12.	Safety Gear (disposable gloves, goggles, face masks)	25
13.	Storage Cabinets for Filaments and Tools	02
14.	Drying Oven for Filament (to keep filament dry)	02
15.	3D Printer Enclosures (for temperature control during printing)	02
16.	Resin tank or vat	04
17.	LCD screen	02
18.	Build plate for MSLA	02
19.	FEP film	04
20.	Nozzle	02

The following conditions must be fulfilled –

- The institute shall not use the same facilities for any other projects/organizations offering a similar course.
- The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facility required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirement of competency standards (CS) provided by SICIP.

For the operation of training course on 3D Printing Technology, the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

The institute will also provide all other hand tools and power tools as per CS for 25 trainees. Also, they will arrange adequate seating arrangement and classroom setup for the 25 trainees.